

segment.) Choose $A_n \in [X]^{<\omega}$ so that some bad sequence in $[X]^{<\omega}$ starts with $A_0, \dots, A_n$ and $|A_n|$ is as small as possible.

Clearly, $(A_n)_{n \in \mathbb{N}}$ is a bad sequence in $[X]^{<\omega}$; in particular, $A_n \neq \emptyset$ for all n . For each n pick an element $a_n \in A_n$ and set $B_n := A_n \setminus \{a_n\}$.

By Corollary 12.1.2, the sequence $(a_n)_{n \in \mathbb{N}}$ has an infinite increasing subsequence $(a_{n_i})_{i \in \mathbb{N}}$. By the minimal choice of A_{n_0} , the sequence

$$A_0, \dots, A_{n_0-1}, B_{n_0}, B_{n_1}, B_{n_2}, \dots$$

is good; consider a good pair. Since $(A_n)_{n \in \mathbb{N}}$ is bad, this pair cannot have the form (A_i, A_j) or (A_i, B_j) , as $B_j \leq A_j$. So it has the form (B_i, B_j) . Extending the injection $B_i \rightarrow B_j$ by $a_i \mapsto a_j$, we deduce again that (A_i, A_j) is good, a contradiction. $\square$

12.2 The graph minor theorem for trees

The graph minor theorem can be expressed by saying that the finite graphs are well-quasi-ordered by the minor relation $\preceq$. Indeed, by Proposition 12.1.1 and the obvious fact that no strictly descending sequence of minors can be infinite, being well-quasi-ordered is equivalent to the non-existence of an infinite antichain, the formulation used earlier.

In this section, we prove a strong version of the graph minor theorem for trees:

Theorem 12.2.1. (Kruskal 1960)

[12.7.1]

The finite trees are well-quasi-ordered by the topological minor relation.

We shall base the proof of Theorem 12.2.1 on the following notion of an embedding between rooted trees, which strengthens the usual embedding as a topological minor. Consider two trees T and T' , with roots r and r' say. Let us write $T \leq T'$ if there exists an isomorphism φ , from some subdivision of T to a subtree T'' of T' , that preserves the tree-order on $V(T)$ associated with T and r . (Thus if $x < y$ in T then $\varphi(x) < \varphi(y)$ in T' ; see Fig. 12.2.1.) As one easily checks, this is a quasi-ordering on the class of all rooted trees.

$T \leq T'$

Proof of Theorem 12.2.1. We show that the rooted trees are well-quasi-ordered by the relation $\leq$ defined above; this clearly implies the theorem.

(12.1.3)

Suppose not. To derive a contradiction, we proceed as in the proof of Lemma 12.1.3. Given $n \in \mathbb{N}$, assume inductively that we have chosen a sequence $T_0, \dots, T_{n-1}$ of rooted trees such that some bad sequence of rooted trees starts with this sequence. Choose as T_n a minimum-order

T_n

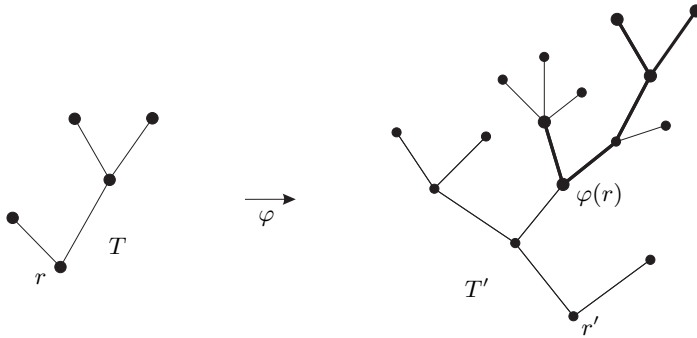


Fig. 12.2.1. An embedding of T in T' showing that $T \leq T'$

rooted tree such that some bad sequence starts with $T_0, \dots, T_n$. For each $n \in \mathbb{N}$, denote the root of T_n by r_n .

Clearly, $(T_n)_{n \in \mathbb{N}}$ is a bad sequence. For each n , let A_n denote the set of components of $T_n - r_n$, made into rooted trees by choosing the neighbours of r_n as their roots. Note that the tree-order of these trees is that induced by T_n . Let us prove that the set $A := \bigcup_{n \in \mathbb{N}} A_n$ of all these trees is well-quasi-ordered.

Let $(T^k)_{k \in \mathbb{N}}$ be any sequence of trees in A . For every $k \in \mathbb{N}$ choose an $n = n(k)$ such that $T^k \in A_n$. Pick a k with smallest $n(k)$. Then

$$T_0, \dots, T_{n(k)-1}, T^k, T^{k+1}, \dots$$

is a good sequence, by the minimal choice of T^k and $T^k \subsetneq T_{n(k)}$. Let (T, T') be a good pair of this sequence. Since $(T_n)_{n \in \mathbb{N}}$ is bad, T cannot be among the first $n(k)$ members $T_0, \dots, T_{n(k)-1}$ of our sequence: then T' would be some T^i with $i \geq k$, i.e.

$$T \leq T' = T^i \leq T_{n(i)};$$

since $n(k) \leq n(i)$ by the choice of k , this would make $(T, T_{n(i)})$ a good pair in the bad sequence $(T_n)_{n \in \mathbb{N}}$. Hence (T, T') is a good pair also in $(T^k)_{k \in \mathbb{N}}$, completing the proof that A is well-quasi-ordered.

By Lemma 12.1.3,¹ the sequence $(A_n)_{n \in \mathbb{N}}$ in $[A]^{<\omega}$ has a good pair (A_i, A_j) ; let $f: A_i \rightarrow A_j$ be injective with $T \leq f(T)$ for all $T \in A_i$. Now extend the union of the embeddings $T \rightarrow f(T)$ to a map φ from $V(T_i)$ to $V(T_j)$ by letting $\varphi(r_i) := r_j$. This map φ preserves the tree-order of T_i , and it defines an embedding to show that $T_i \leq T_j$, since the edges $r_i r \in T_i$ map naturally to the paths $r_j T_j \varphi(r)$. Hence (T_i, T_j) is a good pair in our original bad sequence of rooted trees, a contradiction. $\square$

¹ Any readers worried that we might need the lemma for sequences or multisets rather than just sets here, note that isomorphic elements of A_n are not identified: we always have $|A_n| = d(r_n)$.

12.3 Tree-decompositions

Trees are graphs with some very distinctive and fundamental properties; consider Theorem 1.5.1, or the more sophisticated example of Kruskal's theorem. It is therefore legitimate to ask to what degree those properties can be transferred to more general graphs, graphs that are not themselves trees but tree-like in some sense.² In this section, we study a concept of tree-likeness that permits generalizations of all the tree properties referred to above (including Kruskal's theorem), and which plays a crucial role in the proof of the graph minor theorem.

Let G be a graph, T a tree, and let $\mathcal{V} = (V_t)_{t \in T}$ be a family of bags $V_t \subseteq V(G)$ indexed by the nodes t of T . The pair $(T, \mathcal{V})$ is called a *tree-decomposition* of G if it satisfies the following three conditions:

- (T1) $V(G) = \bigcup_{t \in T} V_t$;
- (T2) for every edge $e \in G$ there exists a $t \in T$ such that both ends of e lie in V_t ;
- (T3) $V_{t_1} \cap V_{t_3} \subseteq V_{t_2}$ whenever $t_1, t_2, t_3 \in T$ satisfy $t_2 \in t_1 T t_3$.

Conditions (T1) and (T2) together say that G is the union of the subgraphs $G[V_t]$. We call these subgraphs the *parts* of $(T, \mathcal{V})$, and say that $(T, \mathcal{V})$ is a tree-decomposition of G into the bags V_t or the parts $G[V_t]$. Condition (T3) implies that the bags of $(T, \mathcal{V})$ are organized roughly like a tree (Fig. 12.3.1).

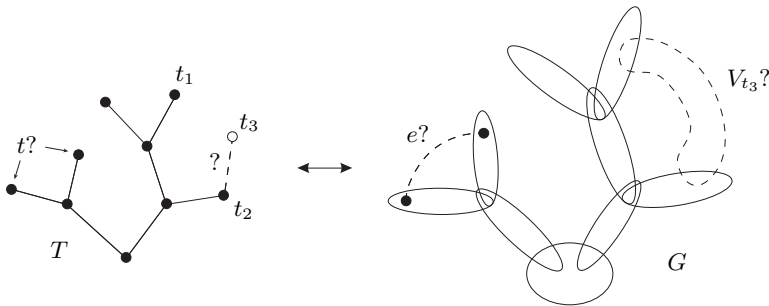


Fig. 12.3.1. Edges and bags ruled out by (T2) and (T3)

Before we discuss the role that tree-decompositions play in the proof of the minor theorem, let us note some of their basic properties. Consider a fixed tree-decomposition $(T, \mathcal{V})$ of G , with $\mathcal{V} = (V_t)_{t \in T}$ as above.

Perhaps the most important feature of a tree-decomposition is that it transfers the separation properties of its tree to the graph decomposed:

² What exactly this 'sense' should be will depend both on the property considered and on its intended application.